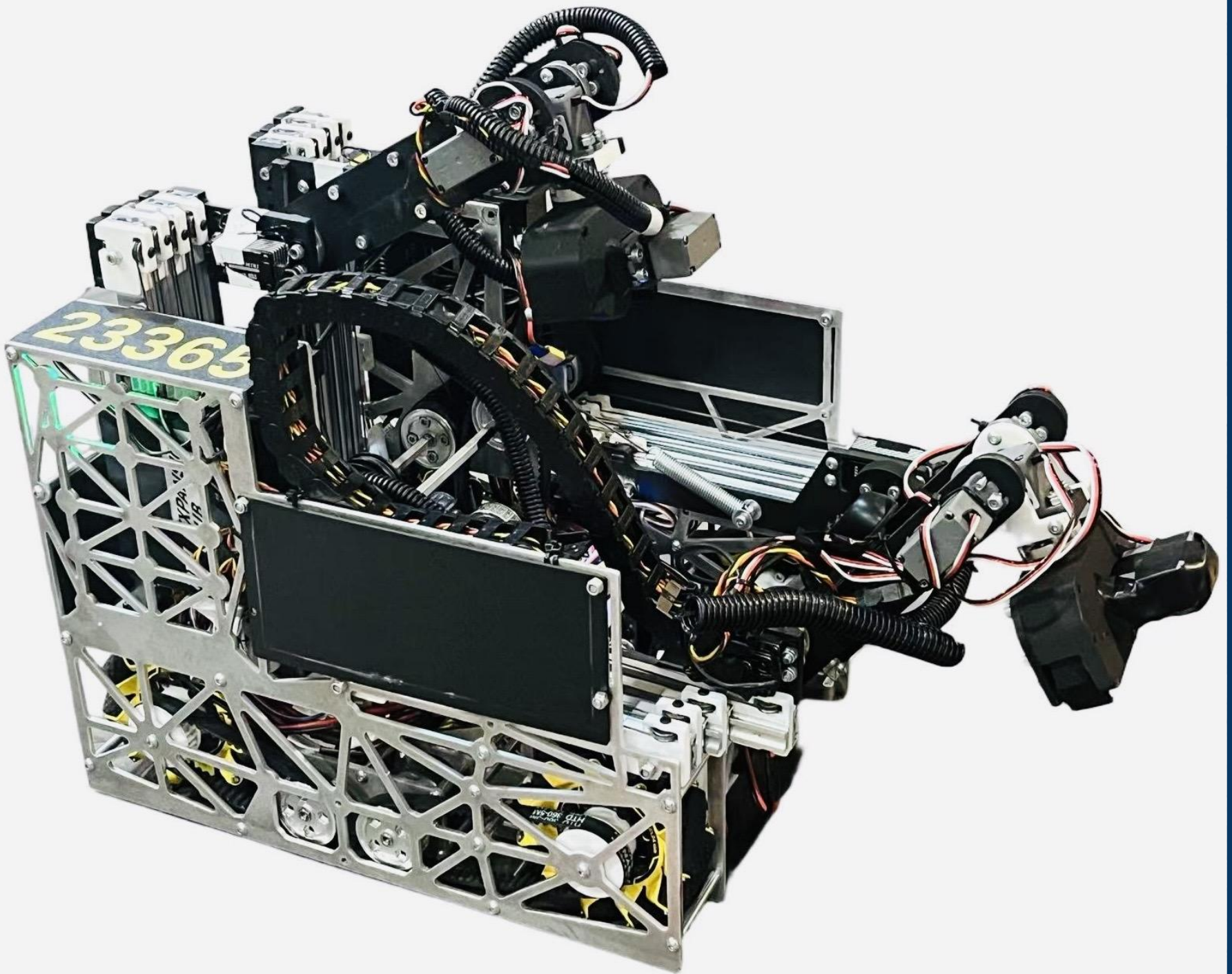


Alpha Robotics Engineering Portfolio

Team 23365



**First Tech Challenge
Team in Socal, Irvine**

Meet the Teams

Start
from
2023

Seven
Team
members
this year



We firmly believe that the foundation of any successful robotics project lies in the the collaborative spirit that drives us to innovate and improve.

Team Member Categories

Team
Captain:
Jason S
11th
grades

Mentor
Coach Zhaolong
Our coach has
been working in
R&D
engineering
companies for
more than 10
years.

CAD and
mechanical
John C 11th
Maximus X
8th
Jason Z 8th

Outreach
Steve L 8th
Robert L
9th

Programming
Jason S
Andy S 9th

Jason Z



Robert L



Maximus X



Jason S



Coach
Zhaolong



Andy S



Steve L



John C



Connections with professions and other teams

Mr Xiao (scientist and nuclear fusion expert from China)

We asked Mr Xiao some questions about our robot's motors selection and vision senses equipment. He gave us some solutions about our old robot's issues.



Team 21980 (Bread Pandas)

We organized a online meeting which asked them about how to improve our drivetrain and linear slides" speed. They told us that in addition to reduce the robot weight, we should use a servo power module which can make motors' speed increase from 435 rpm to >600 rpm.



Game Strategy

Similar with other team, at the beginning we just score the preload specimen. Then push those three sample to the observation zone. Finally we just use our outtake to score the rest 4 specimens and do parking.

During teleop, we basically focused on specimen. (our robot's design is better for specimen) We grab the red/blue sample in the middle. (If there is no red/blue sample or they are too far from us. We will do basket. **We can do both sides during the match.**

During endgame, we just do parking or level 0 hang.



5 specimens
autonomous



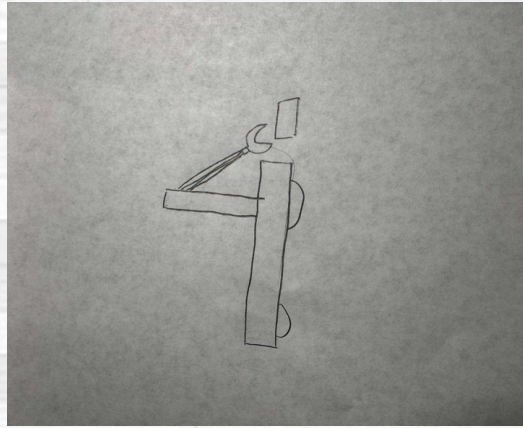
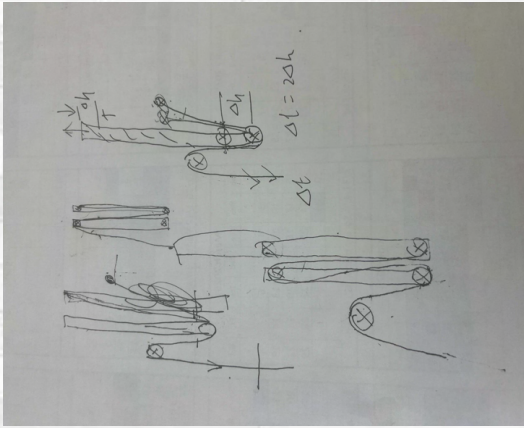
9-11 specimens
15-18 samples



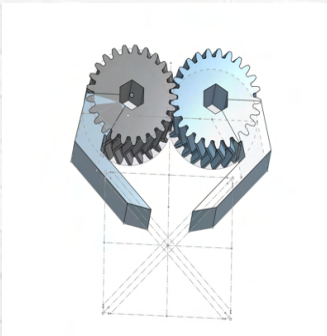
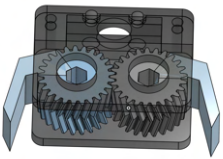
3 points for
parking

Engineering Process

Identifying and Brainstorming



Prototype and Design



Testing and Improve



Goals of our team this year

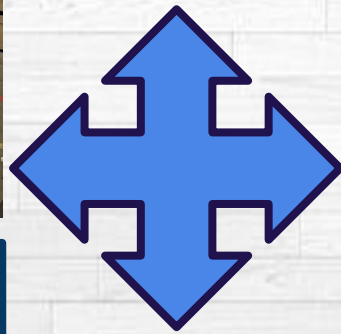
1 Go to state championship this year (already achieved this)

2 Help new members learn engineering and programming skills

3 Become top 100 ftc team in the world

How we made & reach our goals

In order to achieve our goals, we first writing those goals down. We reached our goals by been tolerant and inclusive. Our team went to many robotics and community campaign this year which we studied a lot during these.



We studied from other teams in socal during OC Makers Faire. We found out the gap between those teams and us and foundee out the ways to improve our robot

We saw FTC team Axle Otle during a engineering exhibition. Their robot;s suspension device is really worth learning.

Fundraising

Our mentor's company sponsor \$5000 dollars this year. (27.47%)

We apply funding by sending email to those engineering company. We get \$200 dollars for register. (1.09%)

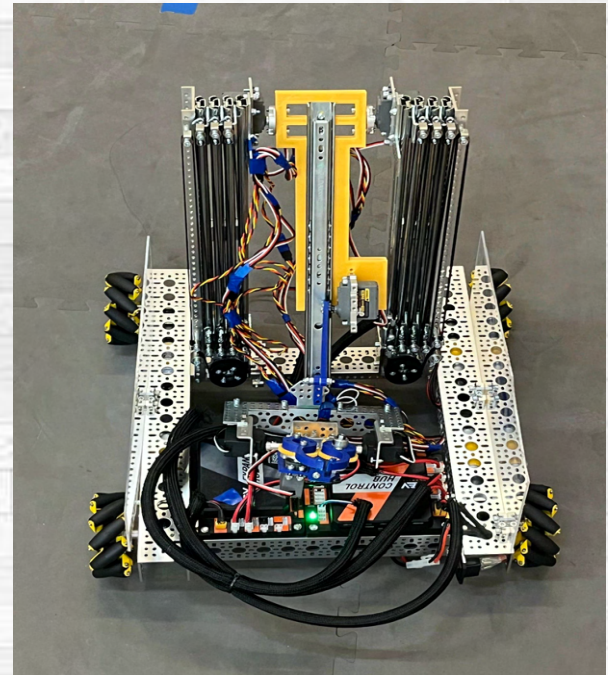
Self-Financed \$13,000 (71.4%)



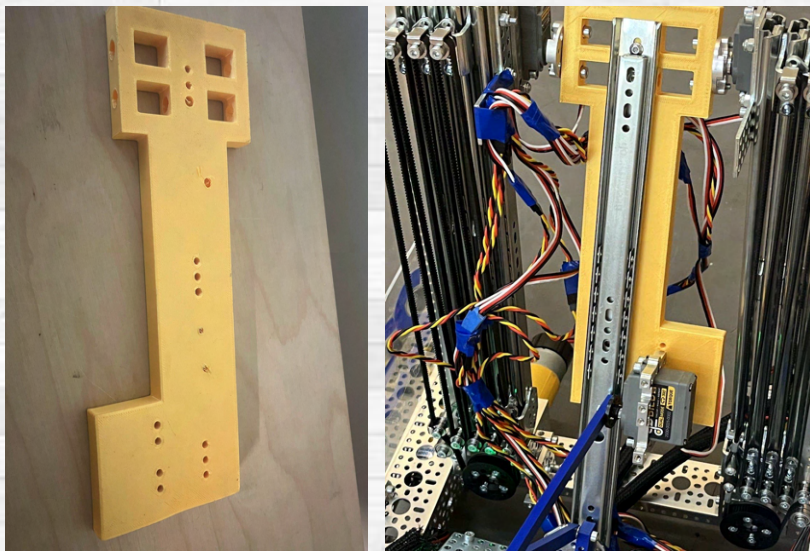
Construction Process 1 (Sample Bot Robot 1)

First Generation: We used a linear slide system, with a vertical slide that can bring things up high and a pivoting slide that can reach things in the submersibles. We designed a linkage brick that connects our reaching slide and claw to the main vertical slide pair. The linkage brick is connected to the vertical slide pair by two servos allowing it to pivot up and down. When the linkage brick wasn't ready yet, our coach helped us create a non custom one using Gobilda parts. (We finished building our first robot before meet 0.)

Picture of Our First Robot



Linkage brick and Our Linear Slides



Issues:

The arm was too heavy so the motor of the linear slides could not lift it up. (weight)

The claw could not grip well (claw design)

The connection lines that connect the control hub and the electronic component were too messy and were easily damaged and disconnected. (cable management)

Couldn't hang(endgame)

Our team's number plate does not meet tournament requirements. (number plate)

Overall, the design failed.

Solutions:

We thought that we needed to build a robot with two sets of linear slide pairs and better cable management. One slide can go horizontal to grab samples. The other slides can go vertically and place the sample into the basket. (higher efficiency and less workload for each part)

Designed new claws and added rubber at the end of each claw for grip

Redesigned number plate and holder for it.

Also new team number cards



Two Stage Linear Slides system on our second robot and we add the rubber at the end of the claw



Our new team number card and it's CAD design cover

Construction Process 2 (Sample Bot Robot 2)

We started building our second robot after meet 0. We thought that scoring samples would be easier and quicker, so our second robot mainly scores samples. We spent nearly 14 hours every week on robotics. After three weeks of hard work, our team finally finished building our second robot. Our second robot uses four 312 mph motors to power our drivetrain and mecanum wheels(can strafe). For our linear slides, we use two 435 mph motors for the vertical linear slide pair and one 1080 mph for the horizontal pair(show in image below). A U-channel is connected to the vertical linear slide pair with a hook(bottom of U-channel) and a specimen claw(top of u channel) that clamps onto specimens. The robot performed well in meetings 2 and 3. Our team created the highest match score with Team Incineers (21982) which is 315 in regional final. We can score an average of 12 samples during teleop in our practice sessions and scored the highest of 14 samples during ILT 1. We were satisfied with the result. However, we know that this isn't the best that we can do. We still need to improve our robot.

Innovative Designs:

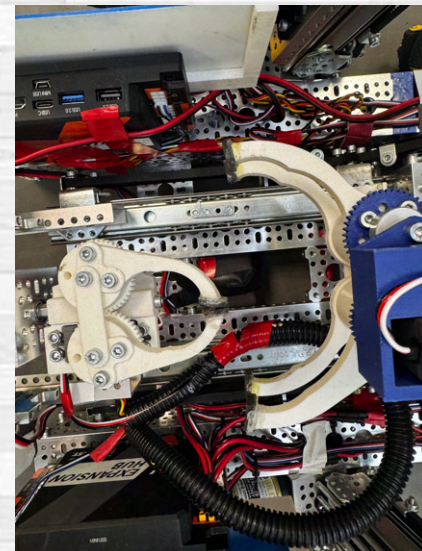
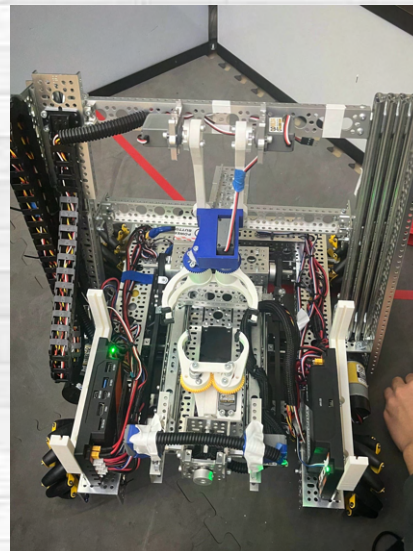
The upper claw

The upper claw, which we use to drop samples, is attached to the inside of the vertical linear slide pair by a two servos and two arms. People usually approach this claw using only two "fingers", but we use 4 as shown in the picture. This is similar to our hands since we have multiple fingers to grab things for more stability and control. So even if the upper two "fingers" don't clamp onto the sample, the lower two "fingers" can prevent the sample from dropping out of the claw and still hold on to it.

The intake claw

Most teams that use passive intake approach the lower claw using super small claws. These small claws are good at grabbing from places where 2+ samples are consecutively touching each other's faces, but they require higher precision in placing the claw which is hard for drivers. Our claw doesn't need the driver to be super precise because it is larger, and this makes our sample pickup time quicker and our drivers can have lesser errors which creates confidence. They work efficiently in almost all cases.

Picture of our second robot



Strengths and Weaknesses

😊 Using two stages linear slides can score the point really fast (efficiency)

😊 claw can precisely pick up samples that you aim for (precision)

Disadvantages :

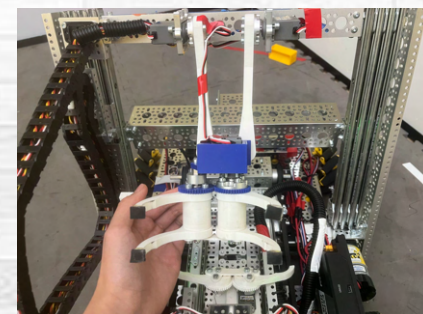
😞 Our robot and linear slides were too heavy so the speed was very low compared to other teams. (Weight and speed)

😞 horizontal claw limited. Cannot reach far enough into the submersible, limited selection of samples

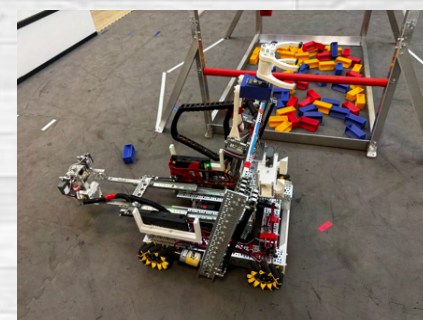
😞 center of mass not low enough, causes little wobbling during moving and reaching slide upwards

😞 robot is too bulky, making moving in small places difficult(size)

Upper Claw



Lower Claw
grabbing
Sample



Construction Process 3 (Specimen Bot Robot #3)

Building Purpose and process:

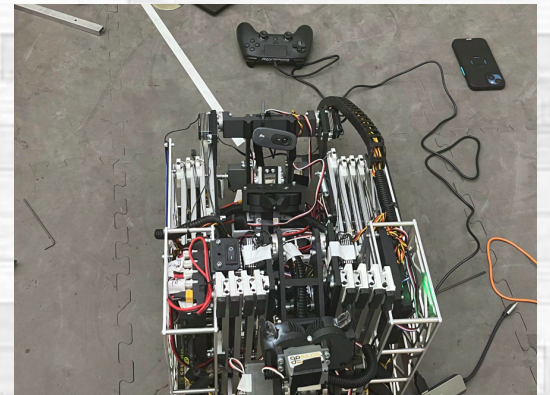
Our third robot, which is the robot we currently use. We started to build our third robot few weeks before the ILT 1 because we realized that our second robot had many weaknesses and will sure to fail in State Championship. Because there were only one more month from State Championship. So every teammates in our team were working really hard. We worked on robotics minimum of 20 hours every week. After 4 weeks of hard working, we finally finished our robot and for the rest 2 weeks, we interpreted our robot and testing autonomous. Although we were really tired, however, we actually felt the significance of building robot and the joy of successfully completing a goal.

Key Features of Our Current Robot (Specs about each part)

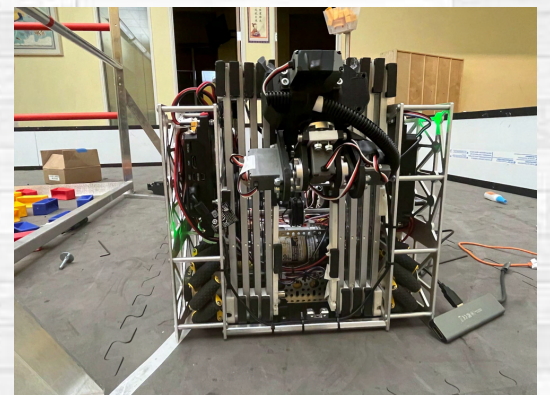
Custom Drivetrain: Our teammate designed a custom drivetrain on onshape which used two inner plates and two outer plates. It is mostly cover by triangle patterns.

Inside the drivetrain: Inside our inner plates, the battery pack is on the front which is covered by frame shown in the top view picture. We attached four 312 RMP motors for our wheels at the very bottom of our drivetrain. Above these four motors, we put two 1150 RPM motors for our outtake linear slides. Above all these gobuilda motors, there is a chamber which connects the scroll wheel and tension line for our outtake linear slides. (shown in the top view picture) These things were separated by some steels and metal pipes. (Back view) In order to avoid the sample fall into our robot, we designed a tilt board to avoid this happens.

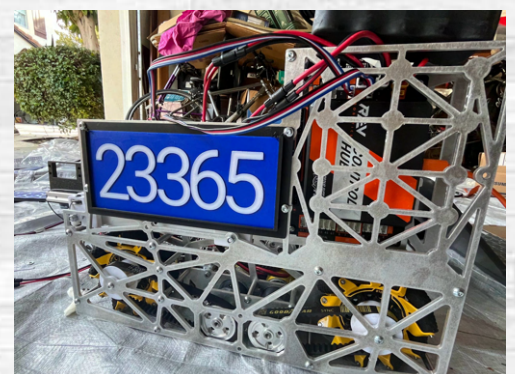
Top View



Back View



Side rear view

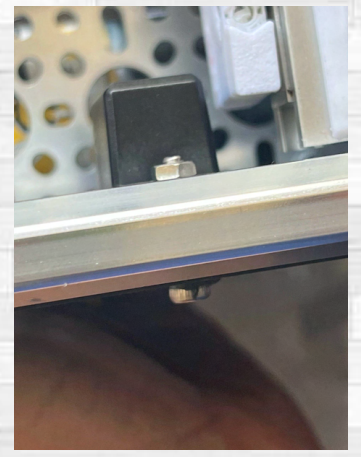


Robot#3 Page 2

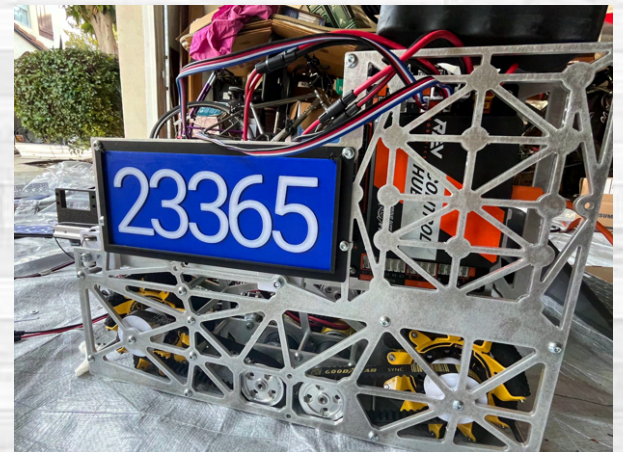
Drivetrain (continue from the front page) : We designed two PLA plates to protect all the motors and wires. This year we used two Odometry for our auto. (One horizontal and one vertical) A really special thing we put on our drivetrain is a slide encoder resetter which helps us test our slides reset position before the match started.

Odometry

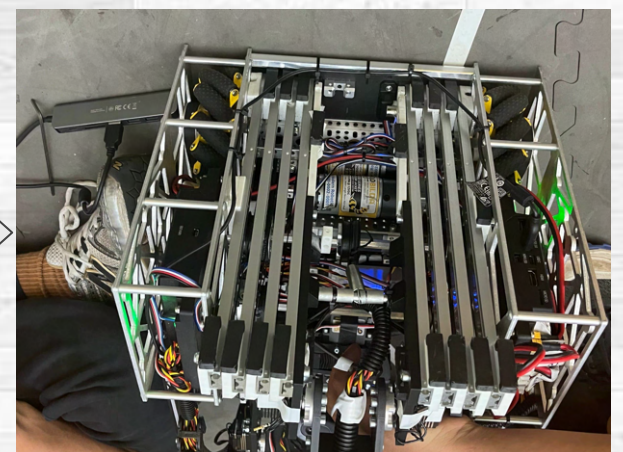
Resetter



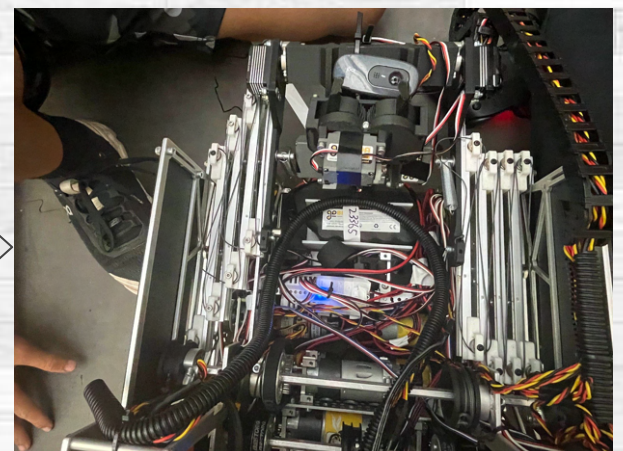
Drivetrain (side): At the side of our drivetrain, we adhesion the team number plates. We put control hub on the left side between two [lates and expansion hub on the right side.



Vertical (outtake) Slides: We used eight 30cm Misumi slides (four on each side) which attached by the linkers. Different with some other teams, we put those slides inside our inner plate. There are two custom designed linkers which we put two Axone servo into it. It connects with our outtake system.

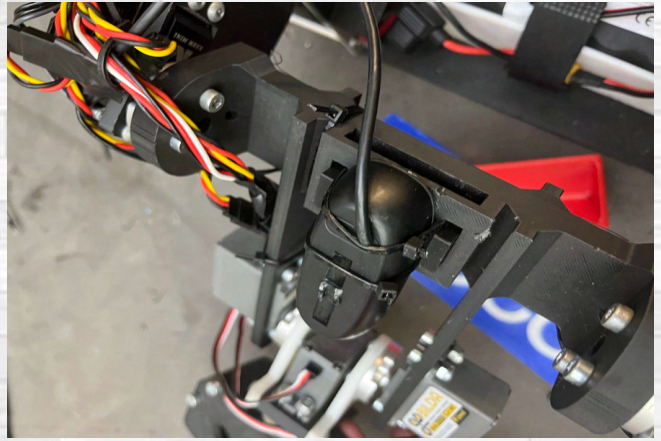


Horizontal (Intake) Slides: We used eight 20cm Misumi slides (four on each side) which attached between the inner plates and outerplates.



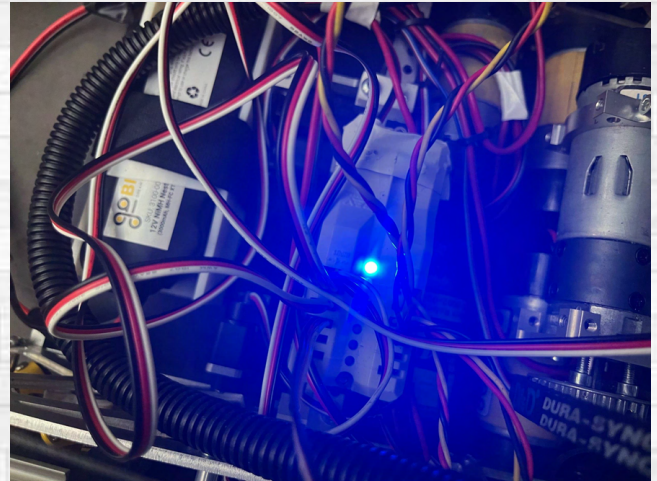
Robot#3 Page 3

Camera: Our claw can fully automatically grab the sample because of the camera's identification.



Servo Power Module

We attached our servo extension wires to the servo power module which increases the speed of the servos and motors. Although this makes our wiring more difficult, however, we thought it is worthy.

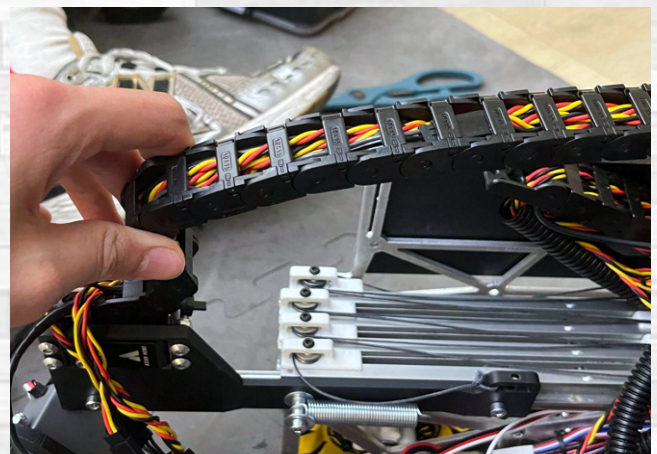
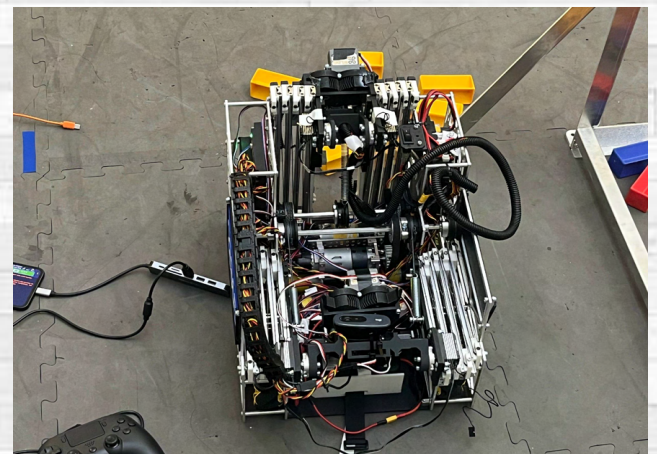


Wiring

outtake: We using plastic pipes to cover all the wires. There is a hole on our outtake system which we can put all wires pass through it. One side of the pipe were tied on the top of our side plate and the other side is attaches on the bottom of outtake.

Intake: Similar methods with outtake.

Drivetrain: We Classifying the wires according to the direction and position of the line. Then use zip tiet to tied them on the plates.

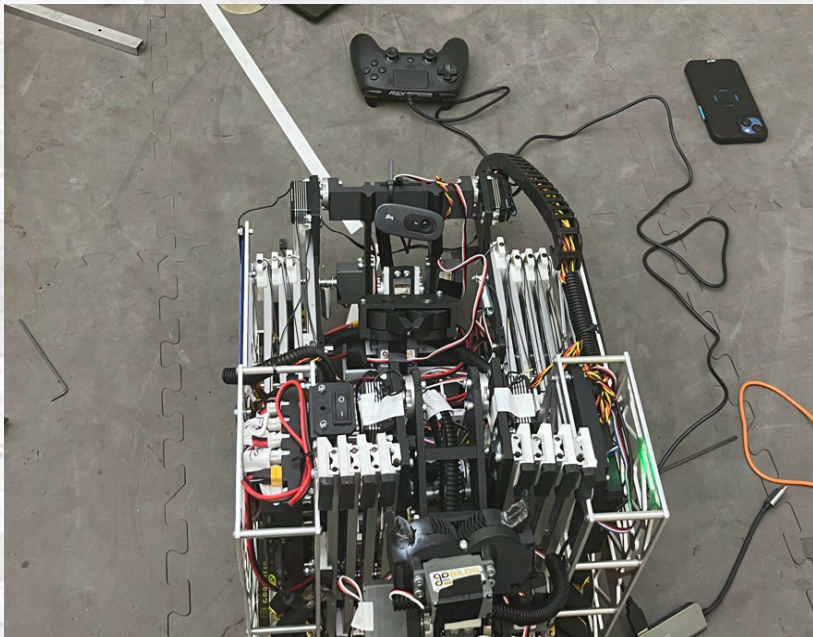
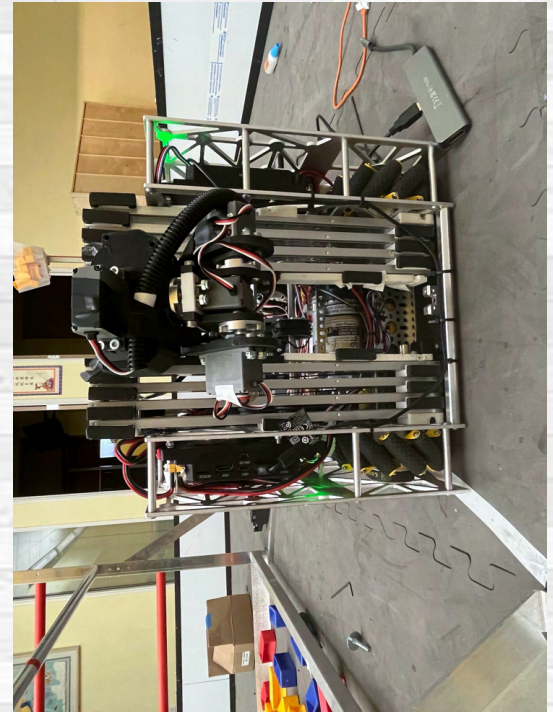


Strengths

- * Really small (14.5 *12 inches) and really fast speed.
- * Fast intake and outtake.
- * Can score both specimen and sample really fast.
- * Three axis intake and camera make our robot grab the sample really efficient.
- * Two-axis outtake make our robot score the sample to the basket or specimen to the chamber really stabilized and efficient.

Robot#3 Page 4 Most Innovative Design

Outtake System: Our outtake is a two-axis system, which we can rotate our arm and claw which different with other teams who can only rotate the arm. We used two Axone mini servos for our outtake arm and three gobuilda torque servos for our claw and upper arm. When our robot want to grab the specimen from the observation zone, our outtake arm first swivel to the back of our machine and grip the specimen. After this, both of our arm and claw will rotate 180 degrees (change to original position) and push to specimen on the higher chamber.



Intake System : Our intake system is a three axis system which we can grab the sample at every side and height. When we want to grab the samples in the middle, we extend the intake slides and the intake system will rotate out. (first rotation) Our claw will rotate any degree or side to best fit the sample placement. (second rotation) After these, the middle part of our intake will rotate down in order to make our claw grab the sample. (third rotation) We designed a linker which connect with our intake slides. We use two Axone mini servos for our back part, two gobuildapeed servo for our middle part and one torque for our claw.

Intake Claw: Our outtake claw design is based on the principle of gears interlocking and rotating. There is a screw hole between two parts of our claw so we can easily attach them. They are small but really sharp and can open really wide so we can grab the sample really tied and accurate.

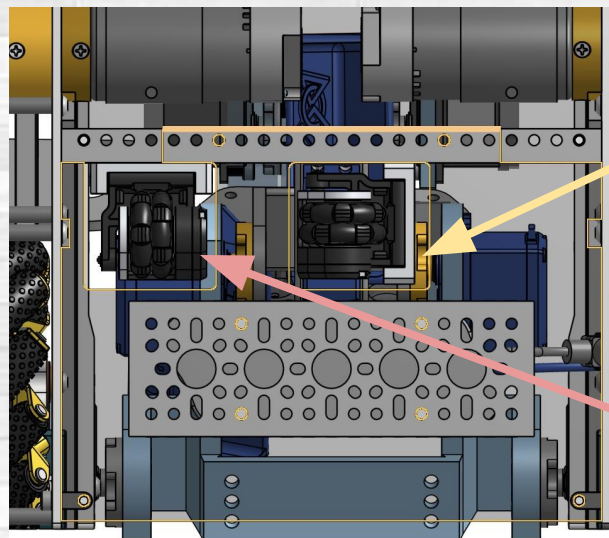
Outtake Claw: Our outtake claw is similar to our outtake claw because they have same design concept. We just made the clamping part shorter and more rounded so it is not easily broken.



Programming

Two Wheel Odometry

In order to achieve a **reliable autonomous** we need to have some sort of localization so that our bot knows where it is on the field. For this bot we decided to use a combination of **two wheel odometry** and **IMU sensor** to provide **accurate localization during autonomous**.



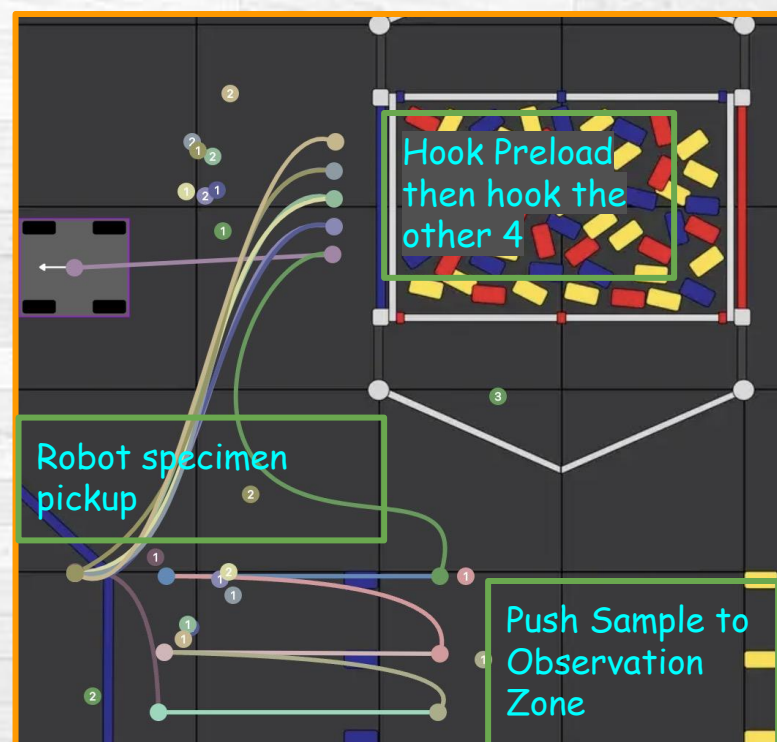
Perpendicular Wheel
Determines Y displacement

IMU sensor
Determines Heading

Parallel Wheel
Determines X displacement

Autonomous Overview

The goal for our autonomous is to score 5 specimens into the high rung, to achieve that we use **Pedro Pathing** a motion planning library developed for FTC which allows us to plan out our path efficiently for a 5 specimen auto. (Path show on the right)



TeleOp Controls

Gamepad 1

Speed Control
Hold for slower speed

Drive Control

Y Stick - Drives forward/backward
X Stick - Strafe left/right

Heading Control

X Stick - Rotates robot
counterclockwise/clockwise

Gamepad 2

Pick up

**Switch Outtake
preset position**

Mode switch

Transfer

**Change Claw
angle**

Intake Slide Control
Y Stick - Extends/Retracts intake slide

Team Spirit

Our FTC robotics team embodies the spirit of collaboration, innovation, and determination. Every challenge we face becomes an opportunity to learn and grow, as each team member contributes their unique skills and perspectives. From brainstorming solutions to fine-tuning our robot, we foster an environment where creativity and teamwork thrive. With every competition, we not only aim to perform at our best but also to inspire and support each other, building lasting friendships and a shared commitment to excellence. The true strength of our team lies in our collective drive to overcome obstacles, adapt to new ideas, and celebrate both our successes and the lessons we learn along the way.

Match Performance and Season Progress

Meet 0

We used our first robot .

2 win 3 lost

We started adapting to the pace of the game and found out the ways to improve our robot

Meet1 and Meet 2

We used our second robot on these 2 matches.

Meet 1: 3 win 2 lost

Meet 2 :4 win 1 lost

We created the event highest score at that time with team 21980(Bread Pandas) 248 points

Meet 3

We won all 5 matches

We learned a lot by talking with other team about their design and game strategy. We recognized we must improve our design

ILT 1

We won all the matches

Our team is the winning alliance.

Design Award 1st place

Advance to higher stage

Created the event highest score with team 21982 which is 315 points.

FTCScout



Team Number

23365

Team Name

Alpha Robotics

Top OPR

146.59

Rookie Year

2023



ftcscout.org



W-L-T: 5-0-0

2.00 RP · 146.59 npOPR · 198.40 npAVG

Dean's List Semifinalist (John Chen), **Dean's List**

Semifinalist (Jason Shen), **Design Award Winner**, **Winning Alliance 1st Pick**

Match

Score

Red Alliance

Blue Alliance



Outreach

Orange County makers faire:

It's a exhibition of engineering and robotics. We attend this event and show visitors our robot and team information. Through this event, our team members saw lots of cool things from other people and learned from colleges, hobbyist, and local businesses.



Network

Youtube links

www.youtube.com/@AlphaRoboticsFTC23365

Instagram name:

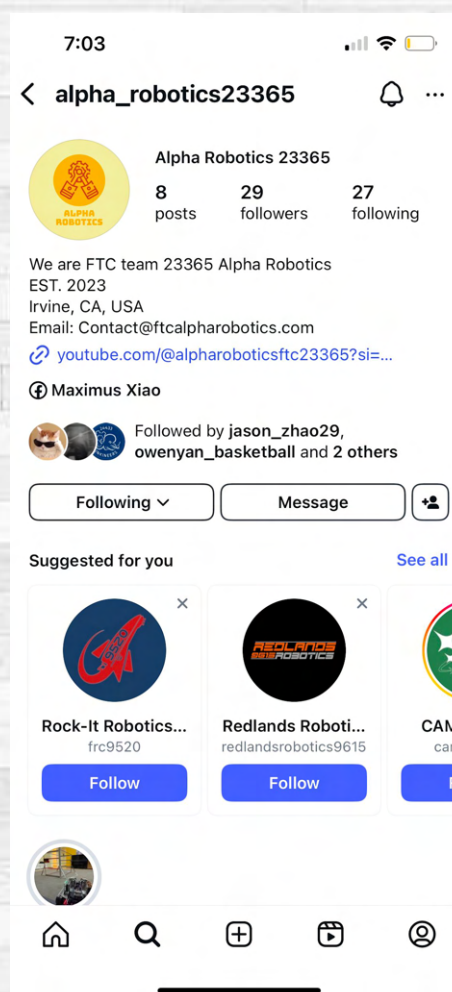
alpha-robotics 23365 (total of 8k views now)

Facebook name: Maximus Xiao

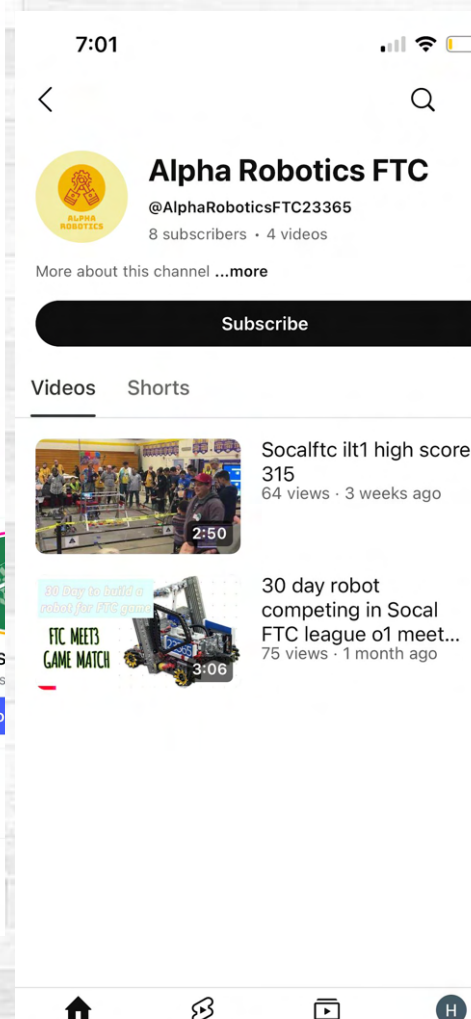
Email:

Contact@ftcalpharoboticsftc23365?=
3365?=-

Instagram



Youtube



Other events we attend

Three scrimmage matches

#1 Held by Axle Otis (Jan 18th)

#2 Held by Watermelon Why (February 7th)

#3 Held by TechAttack (February 22nd)

Future Plans

We will continue to participate in the FTC competition and encourage other teenagers to become a member of the community. We want to go to worlds next year and make our team consistently go to worlds all the years after, inspiring hundreds of teenagers. We argued that experience and spirit of perseverance will directly make us success.

Impressions and Conclusions

Our Impressions

All of us learned and improved a lot during this year's work and competition. Our team's core strength lies in our unwavering perseverance and relentless drive for innovation. We face each obstacle with a determination to push through, knowing that every setback is just another opportunity to improve. Whether it's overcoming technical challenges or refining our designs, we approach each task with creative thinking and an open mind, always eager to try new solutions.

Conclusions:

In conclusion, our robotics team's engineering portfolio reflects not only the technical expertise and innovation we bring to every project, but also our commitment to continuous learning and improvement. Through collaborative problem-solving, creative design, and persistent iteration, we've transformed challenges into opportunities, pushing the boundaries of what we can achieve. This portfolio stands as a testament to the hard work, resilience, and teamwork that have driven our success.

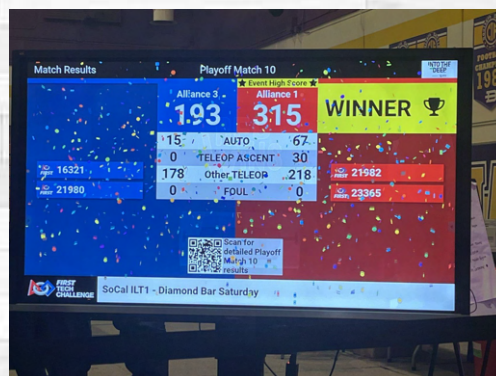


Team motto: Always remember to fall asleep with a dream and wake u with a purpose.

Rescruit for new team members

We will continue to participate in the FTC competition and encourage other teenagers to become a member of the community.

We will recruit new team members by posting posters in our school



Pictures of our team during ILT